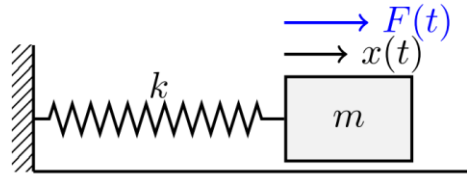
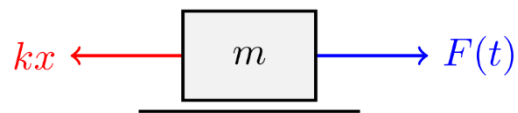


Case 01 – Mass-Spring System

In this configuration, damping was neglected, leaving only the mass and spring.



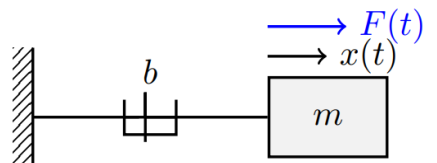
Free Body Diagram



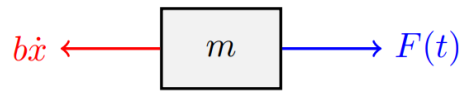
$$G(s) = \frac{1}{ms^2 + k}$$

Case 02 – Mass-Damper System

In this case, the spring was removed and only the damper resisted motion.



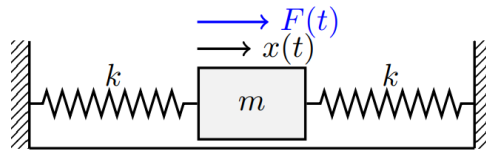
Free Body Diagram



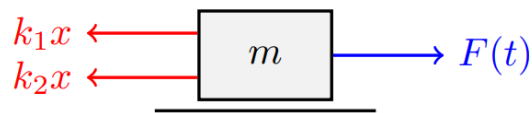
$$G(s) = \frac{1}{ms^2 + bs}$$

Case 03 – Mass with Two Springs

Two identical springs were attached to the mass.



Free Body Diagram



The equivalent stiffness becomes

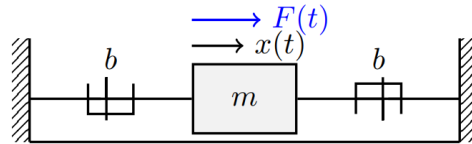
$$k_{eq} = 2k$$

leading to

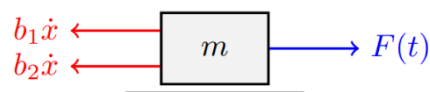
$$G(s) = \frac{1}{ms^2 + 2k}$$

Case 04 – Mass with Two Dampers

Two identical dampers were connected to the mass.



Free Body Diagram



The equivalent damping coefficient becomes

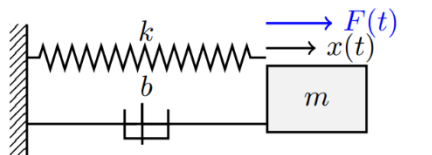
$$b_{eq} = 2b$$

yielding

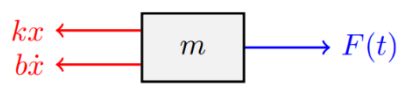
$$G(s) = \frac{1}{ms^2 + 2bs}$$

Case 05 – Spring and Damper on the Same Side

A spring and a damper were connected in parallel between the wall and the mass.



Free Body Diagram

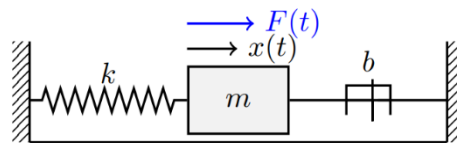


The governing transfer function becomes

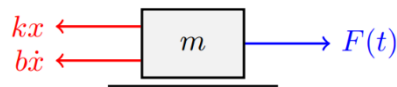
$$G(s) = \frac{1}{ms^2 + bs + k}$$

Case 06 – Spring and Damper on Opposite Sides

The spring and damper were positioned on opposite sides of the mass.



Free Body Diagram



$$G(s) = \frac{1}{ms^2 + bs + k}$$
